Richard Cornelius Suwandi

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Education

The Chinese University of Hong Kong, Shenzhen (CUHK-Shenzhen) — *Sep 2023 – May 2027 (Expected)*
PhD in Computer and Information Engineering
- **Supervisors**: Prof. Feng Yin & Prof. Tsung-Hui Chang (IEEE Fellow)

The Chinese University of Hong Kong, Shenzhen (CUHK-Shenzhen) — *Sep 2019 – May 2023*
BSc in Statistics (Data Science Stream), First-Class Honours

Research Experience

ByteDance (Douyin), Creation & Music Recommendation Team — *Apr 2026 – Jul 2026*
Research Intern, Shenzhen
- Selected for ByteDance's Jindouyun Talent Program (ByteIntern track), a selective recruiting program for high-potential technical talent
- Worked on generative recommendation and LLM4Rec for personalized content generation and discovery
- Applied ideas from CAKE to LLM-driven Bayesian optimization for recommendation and adaptive content selection from feedback

Dria — *Sep 2025 – Feb 2026*
Research Intern, Remote (New York, NY)
- Researched evolutionary coding agents for algorithm discovery and codebase optimization
- Built EvolveBench, a benchmark and execution harness for evaluating coding agents on real GitHub repositories
- Co-developed Kai, an autonomous coding agent for vulnerability discovery, exploit verification, and patch generation
- Built Kai's sandboxed execution environments with Docker, E2B, and Vercel Sandbox
- Shipped Kai as a CLI, FastAPI web UI, and chat-platform integrations

Huawei — *Mar 2025 – Jul 2025*
Research Intern, Shenzhen
- Researched scalable black-box optimization for robust parameter tuning in high-dimensional 5G systems
- Developed ZAP, a gradient-free optimizer for high-dimensional Gaussian process training
- Presented the research outcomes as an oral paper at IEEE ICASSP 2026

Bayesian Learning for Signal Processing (BLSP) Group — *Jun 2021 – May 2023*
Undergraduate Research Assistant (Supervised by Prof. Feng Yin)
- Developed a grid spectral mixture kernel and SLIM-KL, a communication-efficient distributed GP kernel-learning method based on quantized ADMM
- Published the research outcomes at IEEE FUSION 2022 and IEEE TNNLS 2025

Shenzhen Research Institute of Big Data (SRIBD) — *Jun 2021 – May 2023*
Undergraduate Research Assistant (Supervised by Prof. Tsung-Hui Chang)
- Investigated federated matrix factorization for data clustering and recommender systems
- Developed FedMAVg, which combines alternating minimization with model averaging for federated matrix factorization
- Presented the research outcomes as a poster at IEEE ICASSP 2021

Publications

R. C. Suwandi, F. Yin, **"GRAPE: Gradient Refinement and Progress-Aware Exploitation for Query-Efficient High-Dimensional Bayesian Optimization,"** *Preprint*, 2026. Paper Code

Y. Li, R. C. Suwandi, F. Yin, Y. Sun, W. Huang, W. Pu, **"MIMOMamba: From Scalar Duality to Matrix-Valued Attention,"** *International Conference on Machine Learning (ICML)*, 2026. Paper

[ORAL] R. C. Suwandi, F. Yin, T.-H. Chang, **"Breaking the Curse of Dimensionality in Gaussian Process Training With Zeroth-Order Adaptive Perturbation,"** *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2026. Paper Code

R. C. Suwandi, F. Yin, J. Wang, R. Li, T.-H. Chang, S. Theodoridis, **"Adaptive Kernel Design for Bayesian Optimization Is a Piece of CAKE with LLMs,"** *Conference on Neural Information Processing Systems (NeurIPS)*, 2025. Paper Code Poster

R. C. Suwandi, Z. Lin, F. Yin, Z. Wang, S. Theodoridis, **"Sparsity-Aware Distributed Learning for Gaussian Processes with Linear Multiple Kernel,"** *IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*, 2025. Paper Code

R. C. Suwandi, Z. Lin, Y. Sun, Z. Wang, L. Cheng, F. Yin, **"Gaussian Process Regression with Grid Spectral Mixture Kernel: Distributed Learning for Multidimensional Data,"** *International Conference on Information Fusion (FUSION)*, 2022. Paper Code

S. Wang, R. C. Suwandi, T.-H. Chang, **"Demystifying Model Averaging for Communication-Efficient Federated Matrix Factorization,"** *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2021. Paper Poster

Talks & Presentations

- **Sift: A Private Local AI Copilot for Searching, Understanding, and Organizing Your Files**
  Qwen 3.8 Show and Tell (co-organized by Qwen and Alibaba Cloud), Sep 2026. Slides
- **Breaking the Curse of Dimensionality in Gaussian Process Training With Zeroth-Order Adaptive Perturbation**
  [ORAL] IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Barcelona, Spain, May 2026. Slides
- **Adaptive Kernel Design for Bayesian Optimization Is a Piece of CAKE with LLMs**
  Conference on Neural Information Processing Systems (NeurIPS), San Diego, United States, Dec 2025. Slides Poster
- **Adaptive Kernel Design for Bayesian Optimization Is a Piece of CAKE with LLMs**
  [ORAL] Greater Bay Area Graduate Research Forum, Shenzhen, China, Nov 2025. Slides
- **Sparsity-Aware Distributed Learning for Gaussian Processes with Linear Multiple Kernel**
  Doctoral Research and AI Innovation Conference, CUHK-Shenzhen, China, Jul 2025.Slides
- **Gaussian Process Regression with Grid Spectral Mixture Kernel: Distributed Learning for Multidimensional Data**
  International Conference on Information Fusion (FUSION), Linköping, Sweden, Jul 2022. Slides
- **Demystifying Model Averaging for Communication-Efficient Federated Matrix Factorization**
  IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Toronto, Canada (Virtual), Jun 2021.Poster

Honors & Awards

2nd Prize Award, Doctoral Research and AI Conference, CUHK-Shenzhen	2025
IEEE Signal Processing Society Scholarship	2024, 2025
Guangdong Government Outstanding International Student Scholarship	2020, 2021, 2025
Shenzhen Universiade International Scholarship Foundation	2024
School of Science and Engineering Postgraduate Studentship, CUHK-Shenzhen	2023

Undergraduate Research Award, CUHK-Shenzhen	2022, 2023
School of Data Science Dean's List Award, CUHK-Shenzhen	2020, 2021, 2022, 2023
Full Tuition and Accommodation Scholarship, CUHK-Shenzhen	2019
Bronze Medal, Asia International Mathematical Olympiad (AIMO)	2018

Projects

- Kai, autonomous coding agent for verified vulnerability discovery, patching, and software optimization
- EvolveBench,A benchmark and execution harness for evaluating AI-driven research and optimization systems on real GitHub repositories
- OpenEvolve, open-source evolutionary coding agent for algorithm discovery and code optimization
- PlugBO, modular framework for agentic Bayesian optimization
- Awesome BO, a curated repository of Bayesian optimization resources
- PyCaret, low-code machine learning library in Python for automating end-to-end ML workflows
- feature_engine, feature engineering library with scikit-learn-style APIs for production ML pipelines
- pymoo, Python framework for multi-objective optimization

Teaching Experience

Lead Teaching Assistant, CSC6022 Machine Learning, CUHK-Shenzhen	Fall 2026
Lead Teaching Assistant, CIE6007 Machine Learning, CUHK-Shenzhen	Fall 2025
Lead Teaching Assistant, MAT2040 Linear Algebra, CUHK-Shenzhen	Spring 2025
Teaching Assistant, CIE6007 Machine Learning, CUHK-Shenzhen	Fall 2024
Teaching Assistant, MAT2040 Linear Algebra, CUHK-Shenzhen	Fall 2023, Spring 2024

Organizational Experience

Dev Ambassador, Qwen, Alibaba Cloud	2026 – Present
Co-Founder, Institute for AI-Driven Discovery of Algorithms (AIDDA)	2026 – Present
Co-Organizer, AI4Science Community, alphaXiv	2025 – Present
Student Ambassador, CUHK-Shenzhen	2024 – Present
Graduate Student Member, IEEE	2024 – Present
Member, IEEE Young Professionals	2024 – Present
Member, IEEE Signal Processing Society	2024 – Present
Member, IEEE Student Branch, CUHK-Shenzhen	2023 – Present

Services

Journal Reviewer, IEEE Transactions on Signal Processing (TSP)
Journal Reviewer, IEEE Transactions on Neural Networks and Learning Systems (TNNLS)
Conference Reviewer, Neural Information Processing Systems (NeurIPS)
Conference Reviewer, International Conference on Machine Learning (ICML)
Conference Reviewer, International Conference on Learning Representations (ICLR)
Conference Reviewer, IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)
Conference Reviewer, Knowledge Discovery and Data Mining (KDD)